Annotation-Aware WavLM Ensembles for Noise Event Detection in Indian Speech

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***Abstract*—Track 1 asks for the onset and offset of every non-speech noise event in Indian-language speech recordings and scores a submission by event-level F1 plus segment-level Dice. We train frame-level taggers of six designs on 20 ms frames. Each pairs the first four transformer layers of WavLM-Base+ with an optional AudioSet-strong BEATs stream, a bidirectional GRU, and heads for frame presence, clip presence, signed distance to the nearest boundary, and latched start/end states. The test set mixes field recordings with synthetic mixtures, which a waveform-peak test separates exactly, so each population gets its own ensemble and decoding rule. A hysteresis readout and four post-processing stages then model how annotators actually labelled the data. The final system scores 1.5926 on the leaderboard, sixth overall: F1/Dice is 0.494/0.718 on natural clips and 0.993/0.985 on synthetic ones. No external data is used. From the released posteriors the submission rebuilds with event F1 0.9996 against the uploaded file.**

I. Introduction

The VAANI noise-event dataset [1] annotates seven noise categories (animal, vehicle/traffic, baby/child, singing/music, phone/signal/alarm, appliance/machine, human non-speech) in speech from the VAANI corpus [2]. Track 1 is category-agnostic and only the time spans count. The score is $S = F_1 + \text{Dice}$. F_1 is micro-pooled over all clips with greedy one-to-one matching; a match needs both onset and offset within $\max(0.2\, d_{\text{ref}}, 50\,\text{ms})$ of the reference, where d_{ref} is its duration [3]. Dice is computed per clip on a 10 ms grid and averaged over clips. Events are short (median 0.48 s on validation, 14% under 250 ms) and one validation clip in eight contains overlapping events. The tolerance on a 0.48 s event is therefore under 100 ms, and boundary placement dominates the score. We contribute a WavLM [4] frame-tagger ensemble with explicit boundary supervision, route-specific decoding, and an annotation-aware post-pass in which every stage was kept only if it improved a held-out split.

II. Training Data and Preprocessing

Vaani data. Only the official Track 1 release [1] is used (Table I). The human-verified tier (*Gold*, 11,111 clips, 21.85 h) is the core training set. From the unverified tier we keep clips whose longest event lasts at least 1.5 s (*Silver*, 12,014 clips), down-weighted by 0.5. The weakly labelled tier is unused. We split the 2,501 validation clips by speaker (SHA-256 of the speaker id, seed 42) into TUNE (1,257 clips) and DECIDE (1,244 clips), with no shared speakers. TUNE is used for training and as the noise bank below. Thresholds, ensemble composition and every post-processing stage were selected on DECIDE, which no member trains on except v5all (synthetic route only, Table II). **No external dataset is used**; AudioSet [5], [6] enters only through the pretrained BEATs weights (Table III).

Preprocessing. Audio is read with `soundfile`, averaged to mono, resampled to 16 kHz with `librosa`, and stored as int16. Labels are rasterised to 20 ms frames as eight channels: the union of all events (the scored target) and one channel per category. Boundary targets give each frame the signed distance to its nearest onset and offset, clipped to ± 0.5 s. Models see random 5 s crops, each z-normalised.

Augmentation pipeline. We build two offline synthetic stores; there is no other on-the-fly augmentation. *synth_aug* (9,982 clips, seed 0) has two steps. (i) A noise bank is cut from the TUNE synthetic clips as the residual `noisy−clean`, taking each labelled event ± 20 ms and skipping events under 0.15 s. (ii) Each new clip takes one of the quietest Gold hosts of at least 3 s and inserts 1–4 bank events. Each event is resampled by a factor $\mathcal{U}(0.85, 1.18)$, spectrally tilted by a one-pole filter with coefficient $\mathcal{U}(-0.6, 0.6)$, and reversed with probability 0.35. It is placed uniformly at random, avoiding onsets within ± 50 ms of a host event and events longer than $0.8\times$ the host, at an SNR drawn from $\mathcal{U}(-5, 15)$ dB against the host RMS over the same span. Each clip is peak-limited to 0.99. *synth_mask_t* (9,000 clips, seed 5) teaches the long, masking events common in natural recordings. A bank event is looped with 50 ms crossfades to a duration $\mathcal{U}(2, \min(8, 0.85\, d_{\text{host}}))$ s, mixed into a Gold host of at least 4 s at an SNR from $\mathcal{U}(-12, 2)$ dB, and labelled over its full extent.

III. Proposed Method

Architecture (Fig. 1a). We keep the first four of the twelve layers of WavLM-Base+ [4] with the convolutional front-end frozen. One member instead keeps all twelve and learns a softmax mixture of hidden states, initialised at layer 4. The optional second stream is BEATs [7] as fine-tuned on AudioSet-strong labels by PretrainedSED [8]. It supplies a 768-d embedding plus 447 class logits per 40 ms frame, which

Datathon@IndoML 2026, Track 1 (Noise Event Detection). Code, configurations, checkpoints and cached posteriors: https://github.com/SharadhNaidu/noise-event-detection (release v1.0).

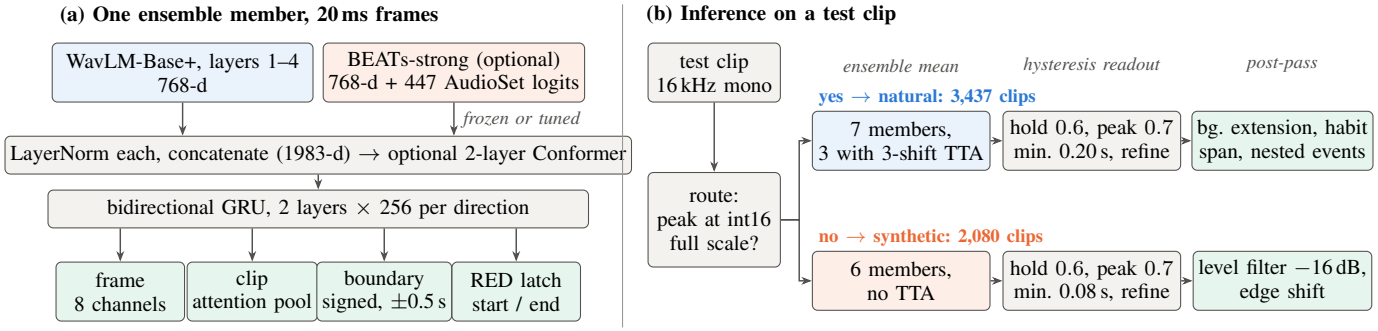


Fig. 1. System overview. (a) The six members differ only in the second stream, the Conformer and the RED head (Table II). (b) Each test clip is routed by its waveform peak to its own ensemble, decoded by hysteresis with boundary-head refinement, and post-processed (Section III).

TABLE I
TRAINING STORES (ALL FROM THE OFFICIAL RELEASE).

Store	Clips	Content and use
Gold	11,111	verified timestamps, 21.85 h; weight 1
Silver	12,014	unverified tier, longest event ≥1.5 s; weight 0.5
TUNE	1,257	speaker-disjoint half of validation
synth_aug	9,982	1–4 bank events in Gold hosts, SNR −5..15 dB
synth_mask_t	9,000	one looped 2–8 s event, SNR −12..2 dB
DECIDE	1,244	held out; model and decoder selection only

we upsample to 20 ms. The stream is either precomputed and frozen or fine-tuned inside the model. Both streams are layer-normalised and concatenated (1983-d), optionally passed through a two-layer Conformer [9] (256-d, 4 heads, kernel 31), and fed to a two-layer BiGRU. Four heads read the GRU states. The frame head outputs eight presence logits. The clip head attention-pools a clip logit. The boundary head regresses signed seconds to the nearest onset and offset. The recurrent event detection (RED) head [10] predicts per-frame start and end probabilities a_t, b_t and decodes presence as the latch $p_t = a_t(1 - p_{t-1}) + (1 - b_t)\, p_{t-1}$, computed with a parallel scan. Where it exists, this latch replaces the frame sigmoid at inference.

Loss. With $\mathcal{L}_{\text{frame}}$ a binary cross-entropy (positive weight 2) over all eight channels, $\mathcal{L}_{\text{clip}}$ the BCE against the crop maximum, and $\mathcal{L}_{\text{bd}}$ a Huber loss on (prediction−target)/50 ms over frames within 0.5 s of a boundary,

$$\mathcal{L} = \mathcal{L}_{\text{frame}} + 0.3\,\mathcal{L}_{\text{clip}} + 0.3\,\mathcal{L}_{\text{bd}} + \left[\mathcal{L}_{\text{pres}}^{\text{focal}} + 100\,(\mathcal{L}_{\text{on}} + \mathcal{L}_{\text{off}})\right]_{\text{RED}}, \tag{1}$$

where the RED terms are a focal ($\gamma = 2$) [11] presence loss and the onset/offset loss of [10]. Clips whose events cover at least 95% of the audio carry no frame contrast and get frame weight 0, except in members trained with whole-clip presence kept (Table II).

Training. We use AdamW [12] (weight decay 0.05) with a one-cycle schedule [13] (15% warm-up). Learning rates are $5{\cdot}10^{-5}$ for WavLM and 10^{-3} for the heads. In v5_S and v12 the BEATs layers use layer-wise decay 0.8 [14] from 10^{-4}. Batches are 6 crops with 3 accumulation steps, gradients are clipped at 5, and training runs in bf16 autocast. A checkpoint is chosen by AP of the union channel minus the capped boundary error on an internal 10% speaker-group split. Table II lists the members; full launch lines are recovered from the configuration stored inside each checkpoint.

TABLE II
ENSEMBLE MEMBERS (ALL WAVLM-BASE+, 5 S CROPS). PARAMS INCLUDE FROZEN WEIGHTS.

Member	Layers	BEATs	Conf.	RED	Ep.	Params	TTA
hon_A	4	–	–	–	8	40.4M	✓
bts_C	4	frozen	2	–	8	43.2M	
bft_S	4	tuned	–	–	8	133.3M	✓
bts_R	4	frozen	2	✓	8	43.2M	
v5_S†	4	tuned, LLRD	–	✓	6	133.3M	✓
v12†	12, mix	tuned, LLRD	–	✓	6	190.0M	

†Whole-clip presence kept. Submitted NAT mean: hon_A, bts_C, bft_S, bts_R, v5_S, and two 12-layer runs (v12, seed 61; v12s62, seed 62). SYN mean: hon_A, bts_C, bft_S, bts_R, and v5all (the v5_S recipe trained on all validation) counted twice.

TABLE III
PRETRAINED MODELS AND RELEASED CHECKPOINTS.

Name	Source, version	Role
WavLM-Base+ [4]	Hugging Face `microsoft/wavlm-base-plus`, revision `4c66d48`	encoder init.
BEATs_strong_1 [7]	PretrainedSED [8] v0.0.1, md5 `0b3d1253`	frame features
6 member checkpoints	GitHub release v1.0 (this work)	inference
Habit classifier	GitHub release v1.0 (this work)	post-pass

Inference and decoding (Fig. 1b). Posteriors are computed on 5 s windows with a 2.5 s hop and averaged where windows overlap. Three members also average three time shifts (zero-padding of 0, 0.64 and 1.28 s). Natural recordings peak at int16 full scale (≥32,765), synthetic mixtures at 0.99 of it; this rule reproduces the natural/synthetic split exactly on validation and divides the test set into 3,437 natural and 2,080 synthetic clips. Decoding uses hysteresis: an event is a maximal run of frames with $p \geq 0.6$ that contains at least one frame with $p \geq 0.7$, so weak bumps are rejected without shortening strong events.

TABLE IV
LEADERBOARD (TEST) SCORES PER SUBMISSION.

Submission	Score	Natural		Synthetic	
		F1	Dice	F1	Dice
Ensemble + readout	1.5831	0.4834	0.7083	–	–
+ bg. extension, syn. shift	1.5852	0.4830	0.7096	0.9925	0.9851
+ habit full-span	1.5902	0.4879	0.7163	0.9925	0.9851
+ type-C nested	1.5907	0.4910	0.7163	0.9925	0.9851
+ TTA, 2nd syn. shift	**1.5926**	**0.4936**	**0.7178**	**0.9929**	0.9845

Final rank: 6. The first file's synthetic split was not reported separately.

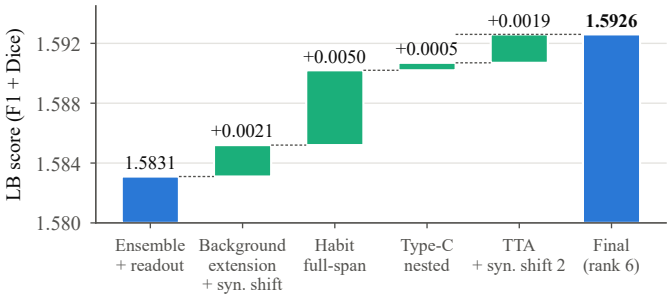


Fig. 2. Leaderboard gain of each post-pass stage over the base ensemble.

Events shorter than 0.2 s (NAT) or 0.08 s (SYN) are dropped. Gaps under 60 ms are merged. Each edge then moves to the median of the boundary-head votes when at least three agree, by at most 0.35 s.

Annotation-aware post-pass. The natural-clip stages follow labelling habits measured on validation. *Background extension* lengthens events of at least 1 s in the animal, singing/music and appliance categories while the union posterior stays above 0.3, and bridges gaps under 0.5 s. A *habit* model adds one clip-length event when annotators would likely have labelled the whole clip and no predicted event already covers 90% of it. It is a histogram gradient-boosted classifier [15] on clip-pooled BEATs tags, duration and decoder state (DECIDE AUC 0.73). *Type-C* adds a category run (score $\geq$0.65, $\geq$0.3 s) nested inside a live event at least twice as long whose dominant category differs. The scorer is category-agnostic and accepts overlapping predictions, so such nested events can still be matched. A separate overlay does the same for human non-speech inside longer events. On synthetic clips we drop events quieter than -16 dB relative to the clip and shift edges by +8 ms (onset) and -6 ms (offset).

IV. Results and Conclusion

Leaderboard. Table IV and Fig. 2 trace the five submissions after the base ensemble. Every post-pass stage raised the score, and the habit stage contributed most (+0.0050, all from natural clips). The natural route is the bottleneck: synthetic clips are near ceiling (F1 0.993), whereas natural F1 is 0.494. On DECIDE, 44% of missed natural references are covered by one prediction that spans several references, 34% are detected with an edge outside tolerance, and 18% are not detected. Leaderboard gains were about 0.46$\times$ the DECIDE estimates.

What did not help. Several measured negative results guided the design. Members built on stronger encoders (WavLM-Large with BEATs, Dasheng [16]) were the best single models on DECIDE (1.4159 vs. 1.3921 for hon_A) but lowered the ensemble by 0.008–0.010 when added to it. Global edge trimming or shifting on natural clips (-0.0015), a Gold-trained edge-correction CNN (-0.0019), duplicate events for merged references (-0.0035 to -0.0050), and test-time shifting of bts_C (-0.0038) all lost on DECIDE and were not submitted.

Reproducibility. Rebuilding the final file from the cached posteriors and scoring it with the official scorer against the uploaded file gives F1 0.9996 and Dice 0.9993; the residue comes from three TTA posterior sets recomputed on a different GPU. Three members (v12, v12s62, v5all) were lost with a cloud instance. Live inference therefore substitutes the epoch-4 snapshot of the 12-layer run and v5_S, and it agrees with the submitted file at F1 0.984 and Dice 0.972 (natural 0.966/0.959, synthetic 0.998/0.993). It processes the 10.8 h test set in 66 min on one laptop RTX 4070.

Conclusion. A compact WavLM ensemble with explicit boundary supervision, together with decoding and post-processing that model the annotators, reached 1.5926 and rank 6 without external data. The remaining error lies in natural recordings, where merged neighbouring events and edge placement account for most misses.

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